BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions)

Maximum Marks: 33

Note: This section consists of 33 part questions and all are to be answered. Each question carries ONE mark. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 85 (Regular & Private)

Q1. Select the most appropriate answer for each from the given options:

Q1. The S.I unit of angular momentum is:

(A) J−s
(B) J/s
(C) S/J
(D) $J-s^2$

Q2. If the velocity of a body is doubled and the mass is reduced to one fourth, the kinetic energy will be:

(A) doubled
(B) unchanged
(C) halved
(D) four fold

Q3. The range of audible sound is:

(A) 1 Hz to 10 Hz
(B) 20 Hz to 20,000 Hz
(C) 21,000 Hz to 24,000 Hz
(D) 25,000 Hz and onwards

Q4. Bragg's law is:

(A) $2d\sin\theta = m\lambda$
(B) $2m\lambda = d\sin\theta$
(C) $d\sin\theta = m\lambda$
(D) $d\cos\theta = 2\lambda$

Q5. If $\vec{A}.\vec{B} = 0$, $\vec{A}\times\vec{B} = 0$ and $\vec{A} \neq \vec{0}$ vector, then $\vec{B}$ is:

(A) zero vector
(B) equal to $\vec{A}$
(C) perpendicular to $\vec{A}$
(D) not parallel to $\vec{A}$

Q6. An object is falling through a viscous fluid with terminal velocity. Its velocity will be:

(A) zero
(B) decreased
(C) increased
(D) constant

Q7. The angle between centripetal and tengential accelerations in circular motion is:

(A) 180°
(B) 90°
(C) 0°
(D) 45°

Q8. One kilowatt hour is equal to:

(A) 3.6×10^8 J
(B) 3.6×10^6 J
(C) 3.9×10^6 J
(D) 3.6×10^9 J

Q9. The weight of a man is 600N at the earth, his weight on the moon where $g_m = g/6$ will be:

(A) 3600N
(B) 300N
(C) 100N
(D) 600N

Q10. The dot product of force and velocity is:

(A) work
(B) power
(C) momentum
(D) energy

Q11. Huygen's Principle is used to determine:

(A) the speed of light
(B) position of wave front
(C) polarization
(D) refractive index

Q12. If $\vec{F} = 3\hat{i}$ and $\vec{d} = 6\hat{j}$, then work done will be:

(A) 2J
(B) 9J
(C) 22J
(D) 0J

Q13. A truck covers a distance of 360km in 5 hours, its speed will be:

(A) 180 km/h
(B) 360 km/h
(C) 72 km/h
(D) 36 km/h

Q14. The point which describe the motion of the whole system or body is called:

(A) inertia
(B) centre of mass
(C) centre of gravity
(D) moments of inertia

Q15. Wave front near the point source is:

(A) plane
(B) spherical
(C) conical
(D) cylinderical

Q16. The first Muslim scientist who invented intravenous injection:

(A) Al-Beruni
(B) Ibn-ul-Haitham
(C) Yaqoob-al-Kindi
(D) Ibn-e-Sina

Q17. The bending of light around sharp obstacles is called:

(A) Interference
(B) Polarization
(C) Refraction
(D) Diffraction

Q18. $\hat{i}.(\hat{k} \times \hat{j})$ is equal to:

(A) 1
(B) -1
(C) 0
(D) $\hat{i}$

Q19. For oscillatory motion of a simple pendulum, restoring force is:

(A) $mg\sin\theta$
(B) $mg\cos\theta$
(C) $mg\tan\theta$
(D) mg

Q20. Dimension of ratio between angular momentum and linear momentum is:

(A) L^{-1}
(B) ML
(C) L
(D) L^{-2}

Q21. If 'M_o' and 'M_e' are magnifying powers of objective and eyepiece of a compound microscope respectively, then total magnification will be:

(A) $M_o \times M_e$
(B) $M_o - M_e$
(C) $M_o + M_e$
(D) M_o/M_e

Q22. A body pushes a toy car, on a horizontal floor with a force of 10N up to a displacement of 2m, then work done by the gravity on the car is:

(A) 20J
(B) 10J
(C) 5J
(D) 0J

Q23. The acceleration of a projectile at the top of the trajectory will be:

(A) g
(B) ½g
(C) zero
(D) 2g

Q24. If a vector has three rectangular components each equal to 'a' the magnitude of vector will be:

(A) $\sqrt{3}a$
(B) 3a
(C) a^3
(D) $\sqrt{3}a$

Q25. In a spectrometer, parallel beam of light is obtained by:

(A) Beam splitter
(B) Collimator
(C) Diffraction Grating
(D) Prism

Q26. It is not a vector quantity:

(A) Force
(B) Torque
(C) Frequency
(D) Weight

Q27. A ball is dropped from a height of 100m, its acceleration at half of its height will be:

(A) 9.8 m/s^2
(B) 4.9 m/s^2
(C) 10 m/s^2
(D) 5 m/s^2

Q28. If elevator moves upward with the acceleration 'a', the apparent weight of a body of mass 'm' in elevator will be:

(A) $mg - ma$
(B) mg
(C) $mg + ma$
(D) zero

Q29. Pitch of sound depends upon:

(A) Amplitude
(B) Intensity
(C) Frequency
(D) Loudness

Q30. 2nd condition of equilibrium is:

(A) $\Sigma F = 0$
(B) $\Sigma F = 0$
(C) $\Sigma \tau = 0$
(D) $\Sigma a = 0$

Q31. A force acting on a body is perpendicular to its displacement, the work done is equal to:

(A) Positive
(B) Negative
(C) Zero
(D) Infinite

Q32. Distance between two consecutive nodes in a standing wave is:

(A) $\lambda/4$
(B) λ
(C) $\lambda/2$
(D) $\lambda/3$

Q33. The number of significant figures in 2.0305 is:

(A) 4
(B) 3
(C) 5
(D) 2

Section 'B' (Short Answer Questions)

Maximum Marks: 32

> Note: Attempt any FIVE part questions from this Section. All questions carry equal marks. Draw diagrams where necessary.

Q2.

(i) Define torque of a couple. Prove $\tau = Fd$ with the help of a diagram.

(ii) Show that any of the following equations are dimensionally correct: ☆ $s = v_i t + \frac{1}{2}at^2$ ☆ $T = 2\pi\sqrt{(l/g)}$ ☆ $T = 2\pi\sqrt{(m/k)}$

(iii) Two converging lenses of focal length 40cm and 50cm are placed in contact. What is the focal length of this lense combination?

(iv) A cyclist is going up a slope of 30° with a speed of 3.5 m/s. If he stops pedalling, how much distance will he move before coming to rest? (Assume, friction is negligible).

(v) Define intensity of sound and loudness. State Weber Fechner law and formula for intensity level.

(vi) What is the ratio of maximum range and maximum height of a projectile for an angle at which range is maximum.

(vii) At what height from center of the earth, the value of 'g' becomes 36% of its value on the surface of earth?

(viii) Two forces $F_1 = 3\hat{i} + 2\hat{j} - 5\hat{k}$ and $F_2 = \hat{i} + 6\hat{j} + 2\hat{k}$, act on a body which is displaced along a vector $\vec{r} = 4\hat{i} - \hat{j} + 3\hat{k}$. Calculate the work done on the body.

(ix) Show that gravitational field is conservative.

(x) Two cars approaching each other from opposite directions with same speed. The horn of one is blowing with the frequency of 3000 Hz and is heared by the people in the other car with the frequency of 3400 Hz. Find the speed of both cars, if speed of sound in air is 340 ms^{-1}.

(xi) A particle of mass 'm' is located at the vector position '$\vec{r}$' and has a linear momentum vector P. The vectors '$\vec{r}$' and P are non-zero. If the particle moves in xy-plane, prove that L_x = L_y = 0, L_z ≠ 0.

(xii) Why the central ring of Newton's ring is dark? State the reason.

Section 'C' (Detailed Answer Questions)

Maximum Marks: 20

> Note: Answer any TWO questions from this Section. All questions carry equal marks. Draw diagrams, where necessary

Q3.

(a) Two non-rotating spheres of masses m_1 and m_2, initially moving with velocities u_1 and u_2 respectively in one dimensions collide elastically. Derive expression for their final velocities v_1 or v_2 (diagram is not required).

(b) Describe the addition of two vectors by rectangular components method. Draw labelled diagrma.

Q4.

(a) Describe Young's double slit experiment. Derive the formula for fringe spacing.

(b) By using a ray diagram, derive an expression for the centripetal acceleration in terms of linear velocity, angular velocity and time period.

Q5.

(a) What is an Astronomical Telescope? With the help of a ray diagram derive an expression for its magnifying power when it is focused at infinity.

(b) How standing waves are produced in a stretched string? Derive an expression for frequencies when such a string vibrates in one loop, two loops and n loops.